

SUMANTH KUMAR MEESALA

Senior AI Engineer • LLM & Agent Applications • AI-Native Full-Stack • Node.js • React • TypeScript

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EXECUTIVE SUMMARY

I'm a Senior AI Engineer with 11+ years of production experience across enterprise SaaS, geospatial intelligence, and rapid client product builds. Over the last three-plus years I've converged that engineering depth with a distinct AI-native product engineering practice — shipping LLM-powered features, retrieval systems, and bounded autonomous agents on top of typed JavaScript stacks. I'm currently **Senior AI Engineer at Affle**, where a 360° review platform I built is used by **600+ employees across global offices**, and I was part of the team that built **Moose (acquired by Kaluza in 2025)**. I operate Claude Code, Cursor, GitHub Copilot, and Augment Code as primary tools, not novelties, and treat AI agents as engineering teammates with explicit responsibilities, context, and guardrails. I'm most effective where rapid 0 → 1 shipping meets production-grade rigour — AI-native product squads inside scaleups and senior AI / LLM application engineering roles where AI features must be merged with hard system constraints.

WHAT I OPTIMISE FOR

- **Shipping AI features that actually run in production** — not demos. Latency budgets, cost ceilings, graceful failure modes, and observability are part of every LLM design I do.
- **AI-accelerated engineering velocity** — turning Claude Code, Cursor, and similar tools into a force multiplier through context engineering, slash commands, agent guardrails, and team standards (CLAUDE.md / AGENTS.md).
- **Bounded autonomy** — designing agents that do one well-scoped job repeatedly, with auditable side-effects, instead of free-form “do everything” agents that drift in production.
- **Engineering credibility over hype** — my comfort zone is Node.js, React, TypeScript, AWS; AI is layered on, not pasted over.

CORE COMPETENCIES

AI-Native Product Engineering: LLM application architecture, RAG pipelines (chunking, embeddings, retrieval, re-ranking), context engineering, multi-step agentic workflows, multi-agent orchestration, tool / function calling, structured outputs, evals, guardrails, prompt engineering, AI workflow automation

AI Tooling (Daily Driver): Claude Code, Cursor, GitHub Copilot, Augment Code, MCP servers, autonomous coding agents, AI-assisted code review, agent-driven test generation, AI-assisted refactoring at scale

LLM Stack: OpenAI GPT-4 / GPT-4o, Anthropic Claude (Sonnet / Opus), Llama, Hugging Face; LangChain, LangGraph, LlamaIndex, CrewAI; Pinecone, ChromaDB, FAISS, pgvector

Languages: TypeScript, JavaScript (ES6+), Node.js, Python (used for LLM tooling and pipelines), C# (legacy .NET enterprise)

Backend: Node.js, Express, AWS Lambda, Serverless Framework, REST APIs, WebSockets, SSE, JWT, OAuth 2.0, async data pipelines

Frontend: React 18, Redux / Zustand, AngularJS, Tailwind, Material UI, responsive + real-time UIs, design-system thinking

Data: DynamoDB, MongoDB, SQL Server, MySQL; vector stores (Pinecone, Chroma, pgvector, FAISS)

Cloud & DevOps: AWS (Lambda, API Gateway, CloudFormation, Cognito, S3, DynamoDB, EC2, CloudFront), Docker, CI/CD, Git, JIRA

Practices: Rapid prototyping, MVP shipping, system architecture, technical leadership, code review, spec-driven workflows with AI agents, observability for LLM systems

PROFESSIONAL EXPERIENCE

Affle — Senior AI Engineer

Jun 2025 – Present

Full-time from Jun 2026; contract via DashAnalysis Jun 2025 – Jun 2026. Global adtech & consumer-intelligence company; I build and operate production AI systems used across the organisation.

- Own and operate a 360° review platform in production — **now used by 600+ employees across Affle’s global offices** — turning peer, manager, and self feedback into structured, LLM-synthesised review reports.
- Scaled the platform from initial rollout to a global user base with a focus on reliability, consistent structured outputs, evals, and report quality.
- Build and maintain production LLM features on Node.js + AWS with cost, latency, and observability controls.
- Converted from a contract engagement (delivered via DashAnalysis from June 2025) to a full-time Senior AI Engineer role in June 2026, on the strength of delivered production work.

DashAnalysis Pty Ltd — Senior Full-Stack & AI Engineer

Aug 2020 – Jun 2026

Lead engineer for AI capabilities on a multi-tenant analytics platform; co-designed architecture and owned production delivery of AI-driven features.

- Architected and shipped enterprise-grade RAG pipelines combining LangChain and LlamaIndex on a hybrid Node.js / Python services backbone, indexing technical documentation and product data into Pinecone and ChromaDB. Tuned chunking, embedding, retrieval, and re-ranking; ~40% improvement in retrieval efficiency vs. baseline keyword search. Replaced multi-day analyst lookups with sub-minute, source-cited answers.
- Designed and operated a fleet of autonomous AI agents using LangGraph and CrewAI to automate analytics, reporting, and DynamoDB-driven orchestration. Each agent has a narrow role, explicit tools, and idempotent side-effects — so they can be retried, audited, and rolled back.
- Integrated Claude (Sonnet / Opus) and GPT-4o into Node.js + AWS Serverless backends. Built a shared LLM client layer with retry, timeout, structured-output enforcement, prompt-caching, and per-tenant rate limits. Hallucination incidents and per-request cost both dropped meaningfully after rollout.
- Productionised AI-assisted engineering as the team’s default workflow — standardised Claude Code, Cursor, and Augment Code; authored shared CLAUDE.md / AGENTS.md context, custom slash commands, and MCP-style tool integrations; embedded agents into code-review and test-generation loops. Median feature cycle compressed by ~30%.
- Led the React 18 + TypeScript front-end re-architecture, gradually retiring legacy AngularJS modules; built real-time analytics dashboards over WebSocket streams and a typed component library used across product surfaces.
- Co-built **Moose** (beige.tech/solutions/moose) — enterprise product platform on AngularJS + AWS serverless (Lambda, CloudFormation, S3, Cognito, DynamoDB); owned full-stack delivery and infrastructure-as-code rollout. **Acquired by Kaluza in 2025.**
- Designed and built the **360° review platform (qrank.it)** on AngularJS + Node.js + AWS, integrating LLMs to convert structured peer / manager / self feedback into human-readable review reports — later deployed at scale across Affle (600+ employees).
- Delivered bespoke multi-step agentic workflows (LangGraph + Node.js) for SMB / enterprise clients integrating with their APIs, tools, and data stores — bounded autonomy with fixed tool sets, structured outputs, explicit termination, and reversible side-effects. Days of manual back-office work compressed into minutes.

Blaque Fracture Technologies Pty Ltd — Senior Full-Stack Developer

Jul 2019 – Jul 2020

Senior IC role on a small product team delivering B2B web applications across multiple client engagements.

- Translated business requirements into MERN-stack applications (MongoDB, Express, React, Node.js); owned features end-to-end from spec through to production deployment.
- Designed responsive, state-driven React + Redux interfaces with reusable component patterns; standardised form validation, error handling, and loading states across the product surface.
- Built and maintained REST APIs with JWT-based authentication, role-based access control, and audit logging; wrote MongoDB aggregation pipelines and indexes for read-heavy workloads.
- Integrated third-party services (payment, email, file storage) behind clean adapter layers; participated in technical-design reviews, code review, and release cycles.

Archimedes — Software Developer

Nov 2018 – Feb 2019

Short-term engagement focused on shipping secure customer-facing web flows.

- Built React frontends bridged to Node.js + MongoDB backend services; converted design specs into production-ready, accessible components.

- Implemented OAuth 2.0 authentication and authorization (Google / corporate IdP) with secure session handling, refresh-token rotation, and protected routes.
- Wrote modular Express middleware for request validation, rate limiting, and structured error responses; documented API contracts for client teams.

Blockfreight, Inc. — Software Developer

Aug 2018 – Oct 2018

Product engineering on an early-stage logistics platform.

- Engineered single-page applications using React and Redux integrated with RESTful Node.js / Express endpoints backed by MongoDB.
- Built domain list/detail flows, search, and filtering with focus on perceived performance — skeleton loaders, optimistic updates, and paginated lists for large datasets.
- Authored API services with consistent input validation, error contracts, and logging; improved deployment scripts and developer onboarding documentation.

ContentTerra Software Pvt. Ltd. — Software Developer

Feb 2015 – Jun 2018

Three-year tenure delivering .NET enterprise web applications across multiple client engagements; the foundation of my product engineering practice.

- Delivered .NET enterprise web applications using ASP.NET MVC, WCF, Entity Framework, C#, HTML5, and JavaScript across the full SDLC — requirements, design, build, test, release, and support.
- Designed and built data-access layers; optimised SQL Server queries with LINQ, indexed views, and stored procedures to meet enterprise SLA requirements on large transactional datasets.
- Implemented authentication and authorization (Forms / Windows / claims-based) and integrated with third-party SOAP and REST services for upstream and downstream systems.
- Reviewed pull requests, ramped junior team members on the codebase, and maintained engineering standards across release cycles.

FOUNDER — CONTENT FACTORY (INDEPENDENT AI PRODUCT)

Founder, builder & operator · Live in production · 2,200+ YouTube subscribers · 600k+ views in 4 months

An end-to-end AI content factory I founded, designed, built, and operate on my own. It automates the full script → TTS → video → publish pipeline for a Telugu astrology YouTube channel and other content surfaces — every step driven by LLM agents and open-source models, glued together by typed full-stack code.

- **Stack:** Next.js 16 (App Router) + TypeScript · Tailwind / DaisyUI / Framer Motion · Remotion (programmatic video rendering) · Claude Agent SDK (script generation) · Edge TTS + custom Parler-TTS server (Hugging Face open-source model) · PostgreSQL · googleapis (YouTube + Instagram OAuth 2.0) · @react-pdf/renderer · Cheerio + RSS Parser · astronomy-engine + mhah-panchang.
- **What it does:** Multi-channel content generation (Astrology, Current Affairs, Trending) with channel-specific AI personas; persona-driven script generation via Claude; multi-language TTS (English, Hindi, Kannada, Malayalam, Tamil, Telugu); word-level animated subtitles synced to TTS; batch render of all 12 zodiac signs in one run; auto-publish to YouTube and Instagram with auto-generated metadata, playlists, and scheduling.
- **Extras:** Trend discovery (Google Trends / YouTube / News) feeding AI content recommendations; festival calendar (panchangam) with templated content; Vedic Jatakam birth-chart generator (lagna, dasa, dosha, PDF export).
- **Outcome:** A fully AI-generated YouTube channel at **2,200+ subscribers and 600k+ cumulative views in its first four months** — running unattended end-to-end. A working portfolio of AI-native fundamentals: agentic LLM orchestration, open-source model integration, programmatic video pipelines, Postgres-backed workflows, and OAuth-based platform integrations.

ENGINEERING PHILOSOPHY

- AI features ship like every other feature — with tests, observability, and fall-back behaviour.
- Treat agents as engineers with narrow job descriptions: define their tools, inputs, outputs, and stop conditions explicitly.
- Context is a first-class artefact — CLAUDE.md / AGENTS.md, system prompts, and tool docs deserve the same review rigour as code.

- Bias to shipping. The MVP that exists in production teaches more than the perfect MVP that doesn't.
- Respect engineering fundamentals — type safety, observability, and clear ownership beat any model upgrade.

KEY TECHNICAL ACHIEVEMENTS

- Built a **360° review platform now used by 600+ employees across Affle's global offices** — structured, LLM-generated review reports in production.
- Part of the team that built **Moose, acquired by Kaluza in 2025**, on Node.js / AngularJS + AWS serverless.
- **Founded and operate an AI content system at 2,200+ subscribers and 600k+ views in four months** — a fully autonomous, AI-generated video pipeline.
- Improved retrieval efficiency ~40% for an enterprise documentation corpus by replacing keyword search with structure-aware chunking + hybrid LangChain / LlamaIndex retrieval over Pinecone.
- Cut feature delivery time ~30% by standardising Claude Code, Cursor, and Augment Code as the team's default workflow with shared context files, slash commands, and agent guardrails.
- Stabilised hallucination rate and per-request cost in production via a shared LLM client layer with retry, structured-output enforcement, prompt caching, and per-tenant rate limits.

WHAT I'M LOOKING FOR

Senior AI / LLM application engineering, AI-native full-stack, founding AI engineer, or staff/principal-track roles inside Australia's AI-native startups, AI-first SaaS scaleups, enterprise AI teams, and consultancies (open to Melbourne hybrid and remote across Australia). Most interested in: 0 → 1 product surfaces with a credible AI thesis; teams that treat AI engineering as production engineering (evals, observability, latency budgets); and founders / EMs who want an early hire who can ship product, design system architecture, and raise the bar on AI-assisted engineering for the rest of the team.

EDUCATION & LINKS

Master of Information Technology (Data Analytics) — Deakin University, Australia · Jul 2018 – Jul 2020 · WAM 81.25

B.Tech, Computer Science Engineering — CVR College of Engineering · Aug 2011 – May 2015

Portfolio: beige.tech/solutions/moose · qrank.it.com · github.com/sumanth-kumar-meesala · linkedin.com/in/sumanthkumarmeesala